LINGUAFORCE: Benchmarking Agentive Force of Discourse in Dialogues via Unified Psychological Dimensions

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Abstract—Discourse that drives, guides, or restricts a listener's behavior—orders, threats, inducements, guilt-tripping, peer pressure, gaslighting, and verbal abuse—shares a common perlocutionary mechanism, yet existing benchmarks study each strategy in isolation with incompatible labels and scales. We propose LINGUAFORCE, a unified benchmark for the agentive force of discourse in real-world multi-turn dialogues, built on a three-layer framework: (i) a strategy taxonomy of four families and fifteen types integrating compliance-gaining and speech-act theory; (ii) seven universal psychological dimensions grounded in pragmatics and persuasion research, which specialize to the obligation, constraint, value-judgement, and toxicity channels in the present annotation scheme; and (iii) a linguistic-realization layer linking dimensions to surface cues. We release 3,432 re-annotated dialogues with intensity and seven-dimension labels; reliability is verified through consistency with existing labels and a two-annotator agreement study. The benchmark is designed to support both detection and interpretability, since each prediction can be traced back to a structured pressure profile. We define three benchmark tasks, a two-stage dimension-aware pipeline, and cross-domain evaluation on MentalManip, MultiManip, TalkDown, and ToxicChat, supporting zero-shot recognition of unseen manipulation types.

Index Terms—Agentive force, perlocutionary pressure, computational social science, psycholinguistic framework, large language models, manipulation detection.

I. Introduction

Language does more than describe the world: utterances routinely act on their listeners, driving, guiding, or restricting behavior. This perlocutionary pressure appears across a broad range of everyday strategies—a polite request that trades on reciprocity, an offer that reframes a decision, an authority figure's direct command, a threat or warning, social proof and conformity pressure, an emotional appeal, and covert tactics such as gaslighting. These strategies differ in surface form and intent, yet they share a common function: they operate on the listener's decision space, expanding or constraining the options that appear available. This listener-directed capacity is termed the agentive force of discourse.

Current NLP research fragments this range across disconnected communities. Toxicity detection targets explicit aggression but misses polite or altruistic manipulation [1], [2]. Persuasion technique detection focuses on news articles and rhetorical framing [3]. Psychological-manipulation datasets cover a narrow subset of covert strategies such as gaslighting and intimidation and rarely include benign or transparent requests as controls [4]–[6]. Compliance-gaining research in communication studies provides rich taxonomies of persuasive strategies [7]–[9] but has not been operationalized as a unified NLP benchmark with continuous, comparable measurements. Taken together, these gaps leave open three questions that matter for both science and safety: (i) what common structure underlies strategies as different as a polite request, a threat, and gaslighting; (ii) can a single measurement scale make them comparable; and (iii) can a model trained on a unified pressure space recognize manipulation forms it has never seen?

Prior work studies many individual forms of influence, but the gap is not only taxonomic: without a shared scale, it is hard to tell whether two different strategies are genuinely comparable or merely labeled alike by convention. It remains unclear whether a common set of psychological dimensions can measure pressure across transparent, exchange, social-normative, and covert strategies.

To address this focused gap, we introduce LINGUAFORCE, a benchmark for the agentive force of discourse in real-world multi-turn dialogues. The central question is whether the strategy structure of agentive force can be represented by a shared set of measurable dimensions. Generalization to strategies absent from the training data is treated as a consequence to test, rather than as a second definition of the phenomenon. The paper proceeds in four steps: related work establishes the theoretical boundary; the framework defines the strategy and dimension layers; the dataset section describes construction and validation; and the experiments test measurement, structure, and transfer. Our contributions are threefold:

1) A unified three-layer framework. We integrate compliance-gaining theory, speech-act theory, and persuasion-technique taxonomies into a strategy taxonomy of four families and fifteen types, a universal inventory of seven psychological dimensions, and a

linguistic-realization layer that links each dimension to observable surface cues. The framework specializes to the obligation, constraint, value-judgement, and toxicity channels within the same unified annotation scheme as consistency anchors.

2) A new dataset and benchmark. We construct LINGUAFORCE, a 3,432-dialogue dataset with dialogue-level annotations of intensity and seven psychological dimensions; annotation reliability is verified through consistency with existing labels on shared dialogues.
3) An experimental protocol. We define three tasks (detection, type recognition, and intensity prediction), a two-stage dimension-aware pipeline, a zero-shot evaluation against a direct baseline, and research questions on dimension structure and zero-shot generalization to unseen manipulation types.

II. Related Work

This section establishes the conceptual basis for treating agentive force as listener-directed action rather than as a synonym for toxicity or manipulation. It then identifies the specific gap addressed by the benchmark: existing resources lack a common strategy inventory and a comparable pressure measurement.

A. Datasets of Manipulation and Persuasion

Communication research has long studied compliance-gaining: Marwell and Schmitt [7] extracted sixteen strategies and Wiseman and Schenck-Hamlin [8] fourteen, later organized by Cialdini [9] into reciprocity, commitment, social proof, authority, liking, and scarcity. In NLP, SemEval-2023 Task 3 detects persuasion techniques in online news [3], and recent datasets target covert manipulation in dialogues: MentalManip, MultiManip [4], [5], and ReaMent [6]. These resources are valuable but narrow: they either target monological news text or focus on covert and pathological strategies, omit benign and transparent controls (requests, advice, rational persuasion), and use incompatible label spaces. Toxicity detection [1], condescension detection [2], and microaggression work [10] cover the abusive end of this range but treat hostility as the phenomenon rather than as a channel of listener-directed pressure. LINGUAFORCE covers the negative, listener-restricting portion of this range: pressure that drives or constrains a target in the speaker's preferred direction. Transparent requests, advice, and rational persuasion are retained as controls and boundary cases, not claimed as equivalent to covert abuse. Positive influence such as praise or encouragement may also change a listener's decision space, but it is outside the present release and is reserved for a future extension.

B. Speech Acts and Agentive Force

Speech-act theory distinguishes what an utterance says from what it does in interaction [11], [12]. This distinction provides the starting point for agentive force of discourse: the object of measurement is not whether a sentence contains an aggressive word, but whether it is used to alter a listener's available or acceptable actions. The term is also distinct from the speaker-oriented notion of linguistic agency in anthropology [13]; our unit is the pressure exerted on the listener's decision space.

C. Normative Influence and Moral Foundations

Normative influence invokes duties, roles, morality, authority, or group identity to steer behavior. Moral foundations theory [14] (care, fairness, loyalty, authority, sanctity, liberty) grounds our normative family C: C1 (moral appeal) typically loads on care and loyalty, while C4 (authority) loads on authority and liberty, sharpening the boundary between normative pressure and pure threats (D_2).

D. Agentive Force in LLM Safety Evaluation

Beyond detection, agentive force of discourse matters for LLM safety. Current evaluations measure explicit harms such as toxicity, bias, and refusal correctness but rarely score manipulative styles—guilt-heavy advice, authority-laden justifications, false-dilemma framings—even though they can steer users toward harmful decisions in health, finance, and politics. LINGUAFORCE's seven-dimension profile provides a continuous, interpretable measurement applicable to generated text (e.g., scoring D_2 or D_3 with the same rubric used for human dialogue), making the benchmark a dual-purpose instrument: a detection benchmark for human conversation and a safety rubric for auditing model behavior. In linguistic anthropology, Ahearn [13] uses "linguistic agency" for the speaker's socioculturally mediated capacity to act; to avoid conflation with our listener-directed notion, we use agentive force of discourse (equivalently, perlocutionary pressure) throughout.

III. Framework: A Unified Account of agentive force of discourse

This section gives the paper's main conceptual answer: agentive force is a listener-directed, scalar pressure effect with a functional strategy layer, a psychological measurement layer, and a linguistic-cue layer. The three layers separate what a speaker does, how strongly it acts on the listener, and which observable signals support the judgment.

A. Definition and Inclusion Criterion

We define the agentive force of discourse of an utterance or dialogue as its perlocutionary capacity to drive, guide, or restrict a listener's behavior and decision space: the pressure an utterance exerts to shift the listener's acceptable action set toward the speaker's preferred subset, along one or more of seven channels. Three properties matter. Success is not required: perlocutionary force exists even when the listener resists. Force is compositional: an utterance can combine channels (e.g., guilt plus social proof). Force is scalar: strategies differ in intensity, not only in kind.

A discourse has agentive force of discourse iff it (i) targets a recognizable action or decision of the listener and (ii) aims to drive or restrict behavior through pressure, inducement, norm invocation, emotional leverage, deception, or hostility. The taxonomy below operationalizes these mechanisms; it is not a claim that the fifteen listed types exhaust every possible realization. Pure information and emotional venting are excluded and serve as negative examples, making the negative class meaningful—a common weakness in manipulation benchmarks.

B. Three-Layer Architecture

Figure 1 shows the architecture. Layer 1 is the strategy inventory: what the speaker does. Layer 2 is the pressure profile: how much force is applied along seven psychological dimensions. Layer 3 is the cue map: which observable words, constructions, and discourse patterns reveal that force. Layer 1 contains fifteen types in four families, while Layers 2 and 3 support comparison beyond any one type list. Jointly annotating the three layers lets models reason at the intended abstraction level and enables error analysis impossible with binary labels alone.

C. Strategy Taxonomy

Table I presents the taxonomy. The four families organize the current negative-force scope by the dominant mechanism: A (transparent) contains direct, openly stated requests, advice, orders, and reasoning and provides the behavioral baseline; B (exchange) motivates through anticipated benefit or harm; C (social-normative) invokes duties, roles, morality, authority, reciprocity, or group conformity; D (covert) relies on emotional exploitation, deception, false dilemmas, or hostility. These families are descriptive organizing units, not mutually exclusive psychological causes, and one dialogue may receive multiple type labels. The final column in Table I records the dominant mechanism used by each type; these mechanisms are descriptive and do not imply mutually exclusive causes.

D. Universal Psychological Dimensions

Table II defines seven dimensions on a continuous 0–1 scale (with a companion four-level discrete label), at both turn and dialogue level. Within the present negative-force scope, every listed strategy maps to at least one dimension. The seven dimensions are therefore a proposed measurement basis, not a claim of psychological completeness; their usefulness must be judged by reliability, discriminative value, and transfer. The dimensions are defined directly from the mechanisms represented in the taxonomy, so every dialogue can be embedded in the same seven-dimensional space and evaluated with the same rubric.

The reused dialogues therefore double as a validation set: morally coercive dialogues should receive high D_3 and low-to-moderate D_7. We also expect each family to load on a small set of dimensions—D_3 for family C, D_2/D_4 for B, D_5/D_6 for D—examined via per-family profiles and clustering (Section IV) as quality checks and descriptive evidence.

E. Linguistic Realization Layer

Table III links each dimension to typical surface realizations. The layer makes annotation operational (annotators justify scores from a cue checklist, improving agreement), supports error analysis (failures trace to missing cues, cue ambiguity, or long-range context), and connects LINGUAFORCE to explainability: upstream-parser dimension scores (Section V) are directly verbalizable as cue-based rationales.

F. A Worked Annotation Example

Table IV shows a worked annotation. The dialogue mixes moral appeal (C1) and obligation (C2), hence receives a multi-label type annotation; intensity is high and the profile is dominated by D_3 with secondary D_1. The example illustrates why the task is hard: the language is polite (no D_6) and the pressure implicit (low D_7)—the pattern that defeats lexical toxicity classifiers.

G. Dimension Structure and Interdependence

The dimensions are not assumed independent; their correlation structure is an empirical object that we state explicitly so it can be falsified. (i) D_1 should correlate with D_2 (the harder the push, the narrower the options), with transparent A1 as the predicted exception. (ii) D_3 and D_4 should co-occur in family C (guilt and moral appeal are emotionally loaded) but separate in family B (threats are affectively cold). (iii) D_5 should be roughly orthogonal to the others—gaslighting occurs in both polite and hostile registers—so a strong correlation with D_6 or D_7 would indicate annotation bias. (iv) D_7 should correlate negatively with D_5: covert strategies are typically indirect. The statistical analysis (Section IV) tests these via correlation matrices and per-family profiles; deviations are reported and used to revise the guideline rather than discarded.

IV. Dataset Construction and Analysis

A. Design Principles

This section explains what evidence is needed to make the definition measurable: coverage of the intended negative-force scope, comparable labels for every dialogue, and hard negatives that separate pressure from mere toxicity or ordinary requests. Four principles guide construction. (i) Real-world multi-turn dialogues: authentic interaction rather than scripted text, which lacks the fragmentation and pragmatic embedding of real conversation. (ii) Scope coverage: negative-force and non-force dialogue across the 0–5 range, so benign controls

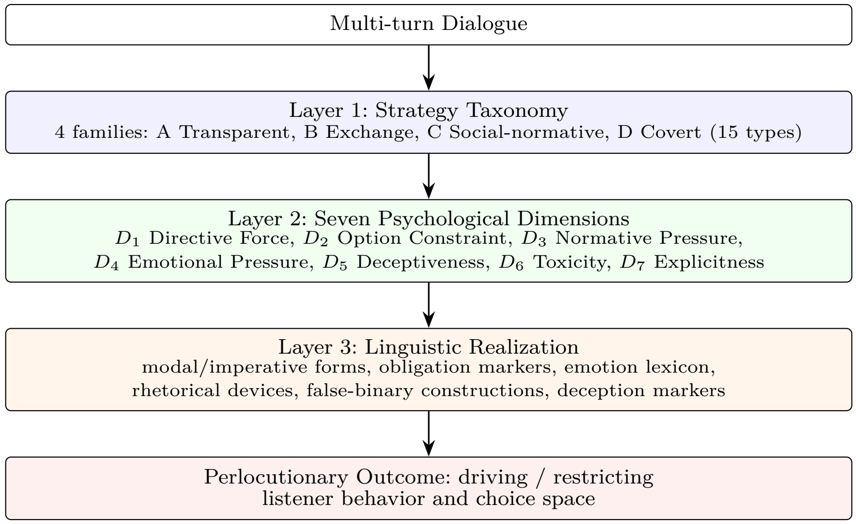

Fig. 1. Three-layer architecture of LINGUAFORCE. The strategy taxonomy defines what the speaker does, the psychological dimensions measure how much pressure is applied, and the linguistic-realization layer links both to surface cues. Normative pressure is expected to be high when the speaker invokes duty, morality, or social roles, while explicitness depends on whether that pressure is stated directly or indirectly.

prevent degenerate models that classify any request as manipulation. (iii) Comparable measurement: every instance receives the same seven-dimension annotation on one scale. (iv) Hard negatives: polite coercion, sarcastic requests, and superficially kind manipulation stress-test lexical shortcuts.

B. Sources and Sampling

LINGUAFORCE re-annotates an existing, de-identified corpus of everyday multi-turn dialogues, retaining its source binary pressure and 0–5 intensity labels for validation. The full release is the 3,432- dialogue training partition; a disjoint 634-dialogue held-out partition is released as a consistency-check set (Section IV-G) and its statistics are reported alongside. Dialogues are already de-identified, so the negative class and the intensity tail arise naturally without new collection.

C. Annotation Protocol

The annotation protocol tests whether the abstract definition can be applied consistently: first identify a listener-directed behavioral target, then the mechanism, and only then its intensity and dimension profile. Annotation follows a three-level decision tree: (1) Does the discourse target a listener action or decision? If no, it is a negative example. (2) Which strategies from Table I are present? (multi-label; a dialogue may combine, e.g., moral appeal and social proof). (3) What are the intensity and the seven dimension scores? Intensity is ordinal 0–5; each dimension receives a continuous 0–1 score and a four-level discrete label (None/Low/Moderate/High). In this release the annotation is produced by an instruction-tuned LLM (Section V) that follows the decision tree and emits a structured JSON profile for each dialogue at temperature 0 for reproducibility. Reliability is validated

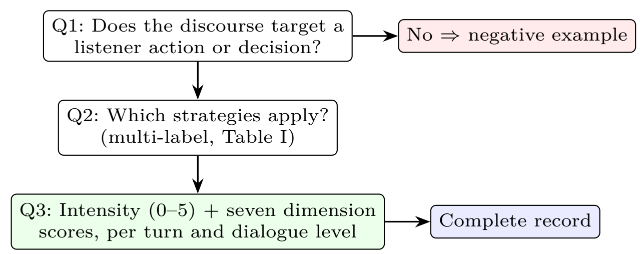

Fig. 2. Three-question annotation decision tree.

through the consistency checks against the existing labels in Section IV-G and through a two-annotator agreement study on a stratified subsample (Section IV-E).

D. Annotation Decision Tree and Negative Cases

To keep the negative class principled, annotation follows the decision tree in Figure 2: the first question filters out ordinary conversation, the second assigns strategy labels, and the third assigns intensity and dimension scores. Negative examples are annotated at the same granularity as positives so the dimension space covers the intended negative-force range. Typical negatives include pure information exchange, emotional venting with no behavioral target, and information-seeking questions. Borderline cases (e.g., advice that is also a subtle request) are resolved by the multi-label rule: assign every applicable type and let dimension scores capture the dominant channel.

E. Quality Control and Consistency Validation

The purpose of quality control is to establish that the proposed dimensions are reproducible measurements rather than labels copied from a single binary judgment. We therefore combine agreement evidence with consistency and monotonicity checks against the source

TABLE I
Strategy taxonomy of LINGUAFORCE: four families and fifteen types. The mechanism column records the dominant pressure channel for each type.

Family	Type	Description and example	Mechanism	Role
A. Transparent	A1 Request/Advice	Low-pressure, intention-transparent proposal. “Could you revise this report by Friday?”	—	benign control
	A2 Directive/Order	Direct instruction with expected compliance. “Send the report before you leave.”	—	transparent pressure
	A3 Rational Persuasion	Argument, evidence, expert opinion. “The data show 30% lower cost; adopt it.”	—	reasoned pressure
B. Exchange	B1 Promise/Inducement	Benefit-driven motivation. “Help me this once and I will return the favor.”	—	incentive pressure
	B2 Threat/Warning	Harm-driven motivation. “If you do not sign, you will face consequences.”	constraint	explicit constraint
C. Social-normative	C1 Moral Appeal/Guilt	Appeals to conscience or moral duty. “A dutiful son would never let his mother suffer.”	normative	normative core
	C2 Obligation/Duty	Invokes role responsibilities. “You are the team leader; it is your duty.”	normative	duty pressure
	C3 Value Judgement/Shaming	Moral labeling of the target. “You are so selfish; you never think of others.”	normative	moral attack
	C4 Authority	Appeals to status, rank, seniority. “The boss decided; why keep arguing?”	—	hierarchical pressure
	C5 Conformity/Social Proof	Appeals to group norms. “Everyone is doing it; why are you special?”	—	peer pressure
	C6 Reciprocity/Debt	Invokes past favors. “I helped you last time; do not refuse this.”	—	debt pressure
D. Covert	D1 Emotional Manipulation	Feigned weakness, pity, or fear to steer behavior. “If you leave too, I do not know what to do.”	—	affective leverage
	D2 Deception/Gaslighting	Distorts facts or memory. “I never said that; you misremember.”	—	epistemic attack
	D3 Logical Trap/False Dilemma	Shrinks the option space. “Either you follow me or we are done.”	constraint	option elimination
	D4 Verbal Abuse/Toxicity	Explicit hostility and disparagement. “You are useless; you cannot even do this.”	hostility	hostile pressure

TABLE II
Seven psychological dimensions used to measure negative agentive force.

Dim.	Definition	Role
D_1 Directive Force	Strength with which the utterance drives a target action	generalization (new)
D_2 Option Constraint	Degree to which the listener's options are narrowed	choice restriction
D_3 Normative Pressure	Pressure via duty, morality, norms, roles	social steering
D_4 Emotional Pressure	Use of guilt, shame, fear, pity as leverage	new
D_5 Deceptiveness	Degree of factual or epistemic distortion	new
D_6 Toxicity	Explicit hostility and abuse	hostile pressure
D_7 Explicitness	Transparency vs. indirectness of the pressure	new

TABLE III
Illustrative surface cues per dimension. Cues are indicative, not sufficient: a cue only counts when the discourse targets the listener's behavior.

Dim.	Typical cues
D_1	imperative and hortative forms; modal “must/should/have to”; performative requests
D_2	“either/or”, ultimatums, no-alternative framings, threats of consequence
D_3	duty and role nouns, moral adjectives, normative “ought”; appeal to what “a good X” would do
D_4	emotion lexicon (guilt, shame, fear, pity), emotional blackmail, self-deprecation
D_5	fact distortions, memory disputes, gaslighting patterns, manufactured consensus
D_6	insults, disparagement, profanity, dehumanizing labels
D_7	direct address and explicit request vs. hints, presuppositions, innuendo

annotations. Because annotation is LLM-produced rather than crowd-sourced, quality control combines measurable consistency with the source corpus's gold labels with a manual re-annotation check. We report, for every dimension, its Spearman correlation with the source binary pressure label and the 0–5 intensity label, and the AUC for separating coercive from non-coercive dialogues (Table VIII); a dimension that fails to track the gold labels is flagged as an unreliable anchor. We additionally check that the predicted intensity is monotonic in the gold label (Figures 5 and 5) and that the inter-dimension correlation matrix shows no pathological coupling (Figures 6 and 6). To probe reliability beyond label consistency, the first author manually annotated a stratified subsample of 120 dialogues (balanced on the gold binary label and spanning all intensity levels), following the same annotation rubric as the automatic pipeline. Agreement with the automatic profiles is high on overall agency strength (Spearman ρ=0.79), and the primary pressure mechanism matches the automatic profile's top two dimensions in 81.2% of the 85 mechanism-bearing cases (excluding the explicitness meta-dimension D_7, which marks directness rather than a specific pressure channel). These results indicate

TABLE IV
A worked annotation example. S = speaker turn; L = listener turn. Type, intensity, and the seven dimension scores are annotated per turn and aggregated at dialogue level.

Turn	Utterance	Type (multi-label)	Dimension profile
S1	"A dutiful son would never let his mother live alone; you are not that kind of person, are you?"	C1 Moral Appeal; C3 Value Judgement	D_3=0.9, D_1=0.7, D_4=0.6, D_7=0.4
L1	"I know, but my job requires me to move..."	— (defensive)	D_1=0.2
S2	"Your mother sacrificed everything for you; it is your turn to repay her."	C2 Obligation; C6 Reciprocity	D_3=0.9, D_2=0.5, D_1=0.8
Dialogue-level			Intensity 4/5; T1 positive; C1+C2+C3+C6

TABLE V
Released LINGUAFORCE sets.

Set	Dialogues	Coercive	Non-coercive
Full release	3,432	1,862 (54.2%)	1,570 (45.8%)
First-release held-out	634	332 (52.4%)	302 (47.6%)

TABLE VI
Inter-annotator agreement on a stratified subsample (n=150; two independent annotators, 149 fully labeled pairs). Binary uses unweighted Cohen's κ; intensity and dimensions use quadratic-weighted κ.

Variable	Cohen's κ	Agreement level
binary (pressure present)	0.49	moderate
intensity (0–5)	0.73	substantial
D_1 directive force	0.55	moderate
D_2 option constraint	0.66	substantial
D_3 normative pressure	0.78	substantial
D_4 emotional pressure	0.60	substantial
D_5 deceptiveness	0.39	fair
D_6 toxicity	0.76	substantial
D_7 explicitness	0.49	moderate

that the seven-dimension profiles are reproducible against independent human judgment.

To further probe reliability beyond the author's spot-check, we ran a two-annotator agreement study on a stratified subsample of 150 dialogues (balanced on the binary label), annotated independently under the same rubric. Table VI reports inter-annotator agreement on the discrete labels: the core intensity label reaches substantial agreement (quadratic-weighted κ=0.73), as do normative pressure (0.78), toxicity (0.76), and option constraint (0.66); binary presence and explicitness reach moderate agreement (0.49), and deceptiveness is the least reliable dimension (0.39), consistent with its subjective, low-frequency nature. The continuous scores inherit the same ordering as the discrete labels, so the reported agreement carries over to the released scores.

F. Statistical Analyses

The analyses below answer one question: do the dimensions behave like a structured measurement of agentive force, rather than seven correlated rephrasings of the same label? We report: (i) per-dimension means and consistency with the gold labels (Table VIII); (ii) the dimension correlation matrix, testing whether D_3–D_4 co-occur while D_5 stays orthogonal (Figure 6); (iii) the separation of coercive and non-coercive dialogues in the seven-dimensional space (Figure 4); and (iv) the monotonicity of predicted intensity in the gold 0–5 label (Figure 5). These analyses also serve as sanity checks against spurious lexical correlations, a failure mode identified in prior manipulation datasets. Type-level analyses on the type-labeled split are reported in Section V-H.

G. First Release and Consistency with Existing Labels

This subsection asks whether the new seven-dimensional profile recovers the known ordering of negative agentive force before the full release is used. Agreement with the source labels is necessary for validity, but by itself it does not establish that the dimensions are useful or causally effective. To validate the framework before scaling annotation, we release a first consistency-check set: the 634-dialogue held-out split of the underlying corpus, re-annotated under the unified seven-dimension scheme. The set contains 332 coercive (52.4%) and 302 non-coercive dialogues, and gold intensity is distributed as 170/54/78/64/107/161 over levels 0–5, so hard negatives and the high-intensity tail are both represented.

Table VII and Figures 3–6 report the consistency of the extracted profiles with the existing labels. All seven dimensions correlate with the source binary pressure label and with the 0–5 intensity label; the strongest channels are option constraint (D_2), directive force (D_1), and emotional pressure (D_4), followed by explicitness (D_7) and normative pressure (D_3); this is consistent with constraint being a core channel of moral-pressure labels. Each dimension separates coercive from non-coercive dialogues with AUC between 0.72 and 0.83. The predicted intensity rises monotonically with the gold label (Spearman ρ=0.66), and the inferred binary decision reaches AUC 0.83 and F1 0.80 at the best threshold, confirming that the unified dimension space reproduces—and can in principle subsume—the prior corpus's own measurement on the shared dialogues. A direct zero-shot judge that bypasses the profile achieves AUC 0.78 and best F1 0.76 on the same dialogues (Section V-H), confirming that the added benefit of the seven-dimension protocol is modest on this already-annotated split but consistent on the harder binary decision. Section IV-H repeats the same analysis on the full 3,432-dialogue corpus.

TABLE VII
Consistency of the seven dimensions with the existing labels on the first release (n=634). ρ: Spearman correlation; AUC: area under the ROC curve for separating coercive from non-coercive dialogues.

Dimension	ρ (binary)	ρ (0–5)	AUC
D_1 Directive force	0.54	0.60	0.81
D_2 Option constraint	0.57	0.63	0.83
D_3 Normative pressure	0.49	0.63	0.78
D_4 Emotional pressure	0.54	0.60	0.81
D_5 Deceptiveness	0.49	0.55	0.77
D_6 Toxicity	0.42	0.35	0.72
D_7 Explicitness	0.53	0.53	0.79
Predicted intensity	–	0.66	0.83

TABLE VIII
Seven-dimension profiles on the full release (n=3,432). Mean score, Spearman correlation ρ with the binary label and the 0–5 intensity label, and AUC for separating coercive from non-coercive dialogues.

Dimension	Mean	ρ (binary)	ρ (0–5)	AUC
D_1 Directive force	0.53	0.51	0.54	0.79
D_2 Option constraint	0.36	0.55	0.58	0.82
D_3 Normative pressure	0.49	0.45	0.55	0.76
D_4 Emotional pressure	0.53	0.57	0.62	0.83
D_5 Deceptiveness	0.12	0.48	0.50	0.77
D_6 Toxicity	0.22	0.40	0.31	0.71
D_7 Explicitness	0.73	0.51	0.49	0.79
Predicted intensity	–	0.63	0.65	0.85

H. Full-Scale Annotation and Dimension Profiles

This subsection asks whether the measurement remains stable when applied to the complete release, including both benign controls and high-intensity negative-force dialogues. Stable profiles and monotonic intensity are the evidence that the proposed abstraction is not an artifact of the initial subset. Having validated the protocol on the held-out split, we scale the seven-dimension annotation to the complete corpus: the full LINGUAFORCE release contains all 3,432 dialogues, of which 1,862 (54.2%) are coercive and 1,570 (45.8%) are non-coercive. Gold intensity is distributed as 986/267/317/337/621/904 over levels 0–5; hard negatives (level 0, 28.7%) and the high-intensity tail (levels 4–5, 44.4%) together dominate the corpus, so the benchmark remains challenging for both the binary and the ordinal decision.

Table VIII summarizes the seven dimensions on the full set. Option constraint (D_2) and emotional pressure (D_4) remain the strongest channels (AUC 0.815 and 0.829), followed by directive force (D_1, 0.792) and explicitness (D_7, 0.788). Deceptiveness (D_5, 0.767) and toxicity (D_6, 0.710) are weaker, and toxicity in particular shows the lowest correlation with the source pressure labels (ρ=0.40 binary, 0.31 intensity): abusive language overlaps with but does not define the coercive core. Since no single channel reaches AUC 0.83 on its own, the coercion decision is not reducible to one surface feature, which motivates the multi-dimensional representation. The predicted overall intensity stays monotonic in the gold 0–5 label across the full corpus (Spearman ρ=0.65; AUC 0.85).

Ethics and data scope. The dataset contains de-identified English dialogues, and the release adds no new personal information or data collection. Detailed provenance, human-subjects considerations, annotation welfare, dual-use risk, cultural scope, and the accompanying datasheet are provided in Appendix A.

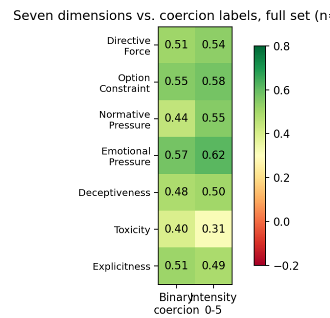

Fig. 3. Seven dimensions versus the binary and 0–5 intensity labels on the full release (n=3,432; Spearman correlation).

V. Experimental Design and Evaluation Protocol

This section turns the conceptual claim into falsifiable tests. T1 asks whether agentive force can be detected, T2 asks whether its strategies can be recognized from the shared profile, and T3 asks whether the same profile orders pressure by intensity. The ablations and cross-domain tests then probe whether the representation is useful beyond the source corpus.

A. Tasks

LINGUAFORCE defines three tasks at increasing granularity. T1 (manipulation detection) is a binary classification of whether a dialogue exerts agentive force beyond the transparent baseline. T2 (type recognition) is a fifteen-way multi-label classification of the strategies in Table I; type labels are released for both the held-out split and the full release. T3 (intensity prediction) is ordinal regression of the 0–5 intensity, evaluated with quadratic weighted kappa (QWK) and accuracy-within-±1.

B. Models and the Two-Stage Pipeline

The pipeline is designed to isolate the proposed intermediate representation: if a profile-based readout matches or improves a direct judge, the gain is evidence that the dimensions expose decision-relevant structure rather than merely renaming the final prediction. Figure 7 shows the two-stage dimension-aware pipeline. An upstream LLM parses a dialogue into an agentive profile, the seven-dimension vector at both turn and dialogue level; a downstream model consumes the profile as structural anchors and predicts T1/T2/T3. This two-stage design decouples dimension extraction from classification. We compare three aggregation modes—turn-level only (T), global only (G), and turn+global (B)—since aggregation granularity is known to affect dimension-based classification.

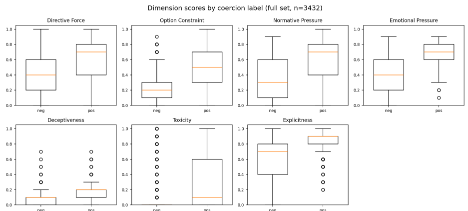

Fig. 4. Dimension scores for coercive versus non-coercive dialogues on the full release (n=3,432).

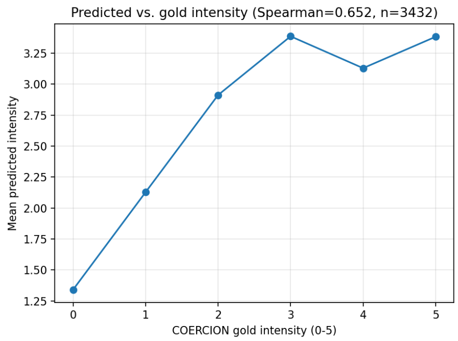

Fig. 5. Predicted intensity as a function of the gold 0–5 intensity label on the full release; points are within-level means.

The pipeline is evaluated zero-shot: a frozen instruction-tuned LLM (DeepSeek) parses each dialogue into the profile, and a direct judge that skips the profile serves as control. Prompt variants include zero-shot, chain-of-thought [15], and self-reflection, following the self-perception approach of MultiManip [5] (Table IX).

C. Experimental Setup

The setup keeps the upstream parser frozen so that the reported comparisons test representation and aggregation choices, not task-specific fine-tuning. All evaluation is zero-shot: a frozen DeepSeek parses each dialogue into the seven-dimension profile and a dialogue-level intensity at temperature 0; T1 and T3 are read off the profile directly, with no fine-tuning. The control is a direct zero-shot judge that skips the profile (dashed path in Fig. 7). Metrics are binary AUC and best F1 over the intensity threshold for T1, and Spearman, QWK, and accuracy-within-±1 for T3. Cross-domain transfer uses the same frozen parser on four public corpora.

D. Research Questions

The experiments are organized around four progressively stronger claims: measurement consistency, complementary structure, transfer to related forms of pressure, and robustness to implementation choices. RQ1 (annotation consistency): Do the seven dimensions align with the source pressure annotations? On the 634-dialogue and full 3,432-dialogue releases (Sections IV-G–IV-H), report per-dimension correlation with the source binary pressure label and 0–5 intensity label, plus each dimension's AUC for separating coercive from non-coercive dialogues.

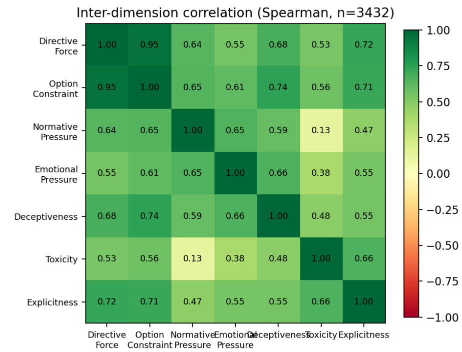

Fig. 6. Inter-dimension correlation matrix on the full release (Spearman).

TABLE IX
Model configuration for the zero-shot evaluation in this release.

Model	Role	Tuning
DeepSeek	upstream dimension parser	frozen
DeepSeek	direct zero-shot judge	frozen

RQ2 (dimension structure): Do the seven dimensions carry distinct, complementary signal? Report per-dimension scores and AUCs, examine the inter-dimension correlation matrix for expected couplings (e.g., D_3–D_4) and unexpected ones (e.g., D_5–D_6), and analyze per-family profiles (Section V-H).

RQ3 (zero-shot generalization): Can dimension conditioning detect unseen manipulation types? Cross-domain transfer to corpora whose types are never seen during training (Section V) yields AUROC 0.742 on MentalManip, 0.706 on MultiManip, and 0.729 on TalkDown. Within-taxonomy leave-one-type-out shows a held-out type is not a separable novel cluster (mean novelty AUROC 0.54, near chance), consistent with types being graded on shared continuous dimensions rather than disjoint clusters. Crucially, the pipeline still flags unseen types as manipulative: predicted intensity separates any-strategy from benign dialogues with AUROC 0.963. The framework thus transfers as a pressure signal, while assigning the specific type label remains open.

RQ4 (engineering conclusions): How do aggregation mode and prompt design interact with the pipeline? Using the same linear readout as Table XII (train full release, held-out evaluation): aggregation (Table X) compares global-only (G), turn-mean (T), and their concatenation (B). Turn-mean pooling collapses toward chance (binary AUC 0.525) because pressure is concentrated in a minority of turns and averaging dilutes it; B performs best (AUC 0.859, Spearman 0.679, QWK 0.680). Prompt variants (Table XI) are comparable (AUC 0.83–0.84), showing robustness to prompt design; CoT marginally helps T1 and self-reflection T3. Remaining grid dimensions (parser-model choice, ICC fidelity) are left to future work.

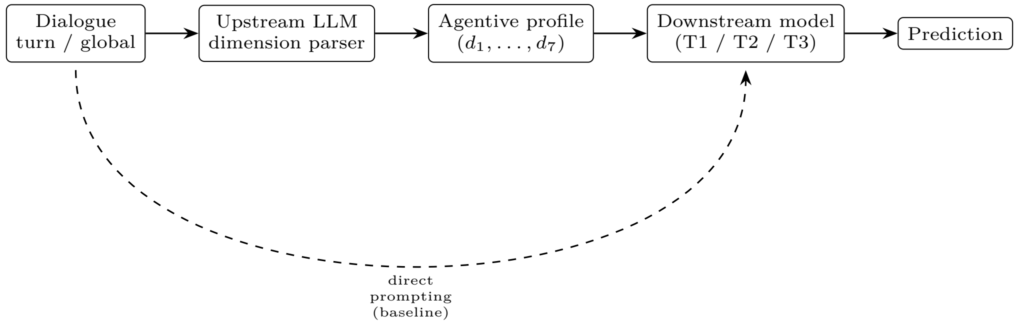

Fig. 7. Two-stage dimension-aware pipeline. The dashed path is the direct prompting baseline used to isolate the effect of dimension anchoring.

TABLE X
Aggregation ablation of the dimension-conditioned pipeline (linear readout; train full release, held-out split). T mean-pools the per-turn profiles over turns; B concatenates the turn-mean and global profiles.

Aggregation	Binary AUC	Intensity ρ	QWK
G (global only)	0.855	0.674	0.667
T (turn mean)	0.525	0.040	0.066
B (turn+global)	0.859	0.679	0.680

TABLE XI
Prompt ablation of the upstream dimension parser (held-out split, n=634). All variants use the same downstream linear readout.

Prompt variant	T1 AUC	T3 Spearman	T3 QWK
Zero-shot	0.832	0.666	0.623
Chain-of-thought	0.839	0.639	0.613
Self-reflection	0.838	0.670	0.630

E. Cross-Domain Evaluation

This subsection tests the scope of the claim: a useful agentive-force profile should transfer to related forms of listener-directed pressure, while remaining distinguishable from toxicity that has no behavioral target. To test transfer and distinctiveness, we evaluate the zero-shot pipeline on existing resources with compatible labels: MentalManip [4] and MultiManip [5] (psychological manipulation), TalkDown [2] (patronizing language), and ToxicChat [1] (toxicity). We report each corpus's standard metric under its own decision rule plus threshold-free AUROC, and use the cross-corpus confusion to identify which constructs are shared with agency and which are distinct. ReaMent [6] and SemEval-2023 Task 3 [3] are left as future work.

F. Dimension Contribution and Anchoring

This subsection asks whether the dimensions carry complementary information about agentive force. A strong result requires more than high aggregate accuracy: removing a dimension should change performance in a way consistent with its proposed mechanism, and a simple readout should retain the signal. Each dimension's contribution is measured by consistency with the gold labels (per-dimension Spearman and AUC, Table VIII) and by a marginal-contribution analysis: a linear readout over the seven scores (logistic for binary, linear for intensity), trained on the full release and evaluated on the held-out split, reaches binary AUC 0.855 and intensity Spearman 0.674—slightly above the LLM's dimension-conditioned readout (AUC 0.831, Spearman 0.665)—so the profiles carry the decision-relevant signal without an LLM. Table XII reports leave-one-dimension-out results: removing toxicity (D_6) degrades detection most (AUC drop 0.008) and normative pressure (D_3) degrades intensity most (Spearman drop 0.021). Notably, D_6 has the weakest univariate correlation yet its removal hurts detection most, indicating non-redundant information (e.g., separating hostile-but-non-coercive dialogues); D_4 is largely redundant with D_3, and the remaining dimensions have small marginal effects, consistent with Figure 6.

TABLE XII
Leave-one-dimension-out ablation with a linear readout over the seven dimension scores (held-out split, n=634). ρ: Spearman correlation of predicted intensity with the gold 0–5 label. The LLM readout is the dimension-conditioned zero-shot reference.

Readout	Binary AUC	Intensity ρ
7-dim linear readout	0.855	0.674
$-D_1$ (directive force)	0.854	0.674
$-D_2$ (option constraint)	0.851	0.668
$-D_3$ (normative pressure)	0.851	0.653
$-D_4$ (emotional pressure)	0.860	0.693
$-D_5$ (deceptiveness)	0.855	0.674
$-D_6$ (toxicity)	0.847	0.669
$-D_7$ (explicitness)	0.853	0.674
LLM readout (reference)	0.831	0.665

G. Dimension-Space Structure

This subsection tests whether the proposed dimensions organize strategy families into interpretable pressure profiles rather than merely reproducing one binary boundary. The expected result is partial overlap with distinct mechanism-specific concentrations, since strategies can share dimensions. To verify that the seven dimensions carve distinct, interpretable regions for the four strategy families, Figure 8 embeds the 3,432 dialogue profiles in two dimensions (t-SNE), colored by the dominant family, together with the 226 dialogues for which no strategy was detected (benign). Families occupy distinguishable regions, and benign dialogues cluster at the low-pressure side. A logistic classifier over the seven raw dimensions separates the five groups with 0.640 accuracy (chance 0.2) in five-fold cross-validation, and each dimension alone separates benign from manipulative dialogues with AUC 0.73–0.94. Figure 9 reports each family's mean dimension profile: the social-normative family peaks on normative and emotional pressure (D_3, D_4), the exchange family on option constraint and toxicity (D_2, D_6), the covert family on emotional pressure (D_4), and benign dialogues are uniformly low across all dimensions.

H. Results on the Full Release

This subsection brings the evidence together. It asks whether the profile improves detection and intensity prediction over a holistic judgment, and whether that improvement survives transfer and fine-grained type recognition. Table XIV and Table XV report the zero-shot results of the dimension-conditioned pipeline on the full 3,432-dialogue release: a dialogue is decoded into the seven-dimension profile (Section V), the profile is mapped to a 0–5 intensity, and T1 and T3 are read off directly. As a control, Table XIV also reports a direct zero-shot judge that skips the dimension profile (dashed path in Fig. 7) on the first-release split. The dimension-conditioned readout reaches binary AUC 0.852 and best F1 0.817 on the full release, exceeding the direct judge (AUC 0.780, best F1 0.755) and the first-release estimate (AUC 0.832, F1 0.795); the fine-grained profile therefore carries information that a single holistic judgment misses, and the estimate is stable as the corpus grows. On the 0–5 intensity scale the readout is well calibrated (QWK 0.595, Spearman 0.651, accuracy within ±1 of 0.692 on the full release), with only a modest drop from the first-release values as hard negatives and the intensity tail are added.

The cross-domain transfer results (Table XVI) support the interpretability and discriminative-validity claims. Under a threshold-free ranking, the zero-shot judge transfers moderately to the psychological- manipulation corpora (AUROC 0.742 on MentalManip, 0.706 on MultiManip) and to patronizing language (AUROC 0.729 on TalkDown), while transfer to generic toxicity is weak (AUROC 0.587 on ToxicChat). This pattern is consistent with the design: manipulation and condescension exert listener-directed pressure and are therefore partially captured by the agency dimensions, whereas generic toxicity is largely distinct from agency. Fixed-threshold macro-F1/F1 values are much lower (e.g., 0.058 on TalkDown) because the intensity scale is calibrated on our annotation scheme and the judge rarely assigns intensity ≥ 3 to dialogues annotated under other corpora's schemes: ranking (AUROC) transfers, absolute calibration does not. Target-calibrated thresholds recover macro-F1 around 0.68–0.74 on the manipulation corpora. T2 type recognition is evaluated with a lightweight one-vs-rest logistic readout over the seven dimension scores, contrasted with a supervised baseline

TABLE XIII
T2 type recognition: a one-vs-rest logistic readout over the seven dimension scores versus fine-tuned RoBERTa-base. "Held-out": trained on the full release (n=3,432), evaluated on the type-labeled held-out split (n=634); "5-fold CV": cross-validation over the full release. F1 is macro/micro at the best threshold.

		Linear readout		RoBERTa-base
Granularity	Protocol	Macro-F1	Micro-F1	Macro/Micro-F1
15 types	Held-out	0.480	0.668	0.544 / 0.731
15 types	5-fold CV	0.477	0.669	0.540 / 0.734
4 families	Held-out	0.772	0.827	0.780 / 0.820
4 families	5-fold CV	0.752	0.820	0.767 / 0.815

TABLE XIV
T1 detection on the full release (n=3,432): the dimension-conditioned readout, versus the same readout on the first-release split (n=634) and a direct zero-shot judge that skips the profile (dashed path in Fig. 7). "Best F1" is the maximum F1 over the intensity threshold.

Metric	Full (n=3,432)	First release	Direct zero-shot
Binary AUC	0.852	0.831	0.780
Best F1 (threshold)	0.817 (3)	0.795 (3)	0.755 (3)
Accuracy @ best F1	0.789	0.760	0.711
Best per-dimension AUC (D_2)	0.829	0.825	—

that fine-tunes RoBERTa-base end-to-end on the dialogue text (Table XIII). We report two protocols for the readout. First, it is trained on the full 3,432-dialogue release and evaluated on the type-labeled held-out split: family-level recognition reaches micro-F1 0.827 and macro-F1 0.772, and type-level recognition reaches micro-F1 0.668 and macro-F1 0.480. Five-fold cross-validation over the full release yields similar estimates (family micro-F1 0.820 and macro-F1 0.752; type micro-F1 0.669 and macro-F1 0.477), so the results are stable as the corpus grows from 634 to 3,432 dialogues. The fine-tuned RoBERTa baseline gives consistent cross-validation estimates as well (type macro-F1 0.540, micro-F1 0.734; family macro-F1 0.767, micro-F1 0.815), confirming that the held-out results are not sensitive to the train/test split. The fine-tuned RoBERTa-base is clearly stronger at type level (macro-F1 0.544, micro-F1 0.731 on the held-out split), confirming that the type labels are learnable by a supervised model; at family level it matches the readout (macro-F1 0.780, micro-F1 0.820), so the dimension readout remains a competitive lightweight and interpretable baseline. Type-level macro-F1 is depressed by three rare types (rational persuasion A_3, authority C_4, deception/gaslighting D_2), which appear in fewer than thirty of the 3,432 dialogues; family-level results are the more reliable target for this corpus, and the full type labels are released for downstream use.

I. Limitations and Threats to Validity

Several limitations should be acknowledged. First, the seven dimensions are a design choice; a finer or coarser decomposition is possible. We mitigate this by reporting consistency against the source pressure labels at both

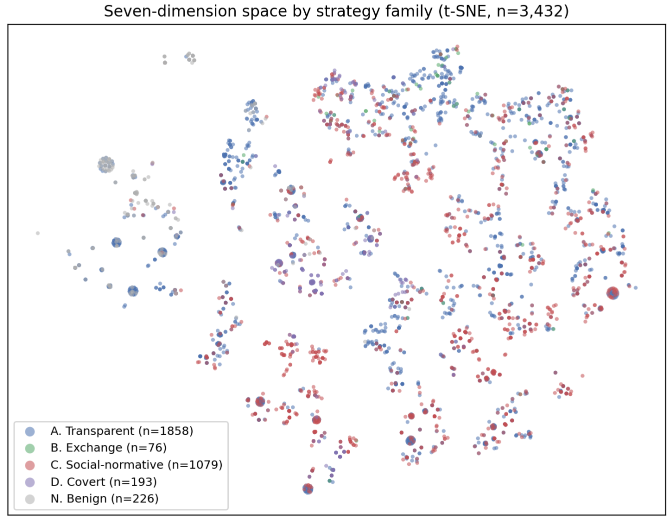

Fig. 8. t-SNE embedding of the seven-dimension profiles (n=3,432), colored by dominant strategy family; the 226 no-strategy dialogues form the benign group.

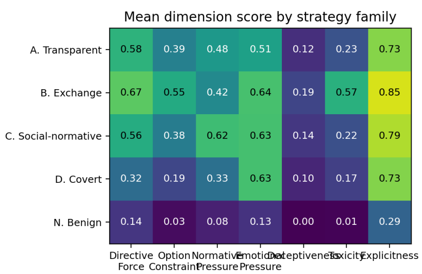

Fig. 9. Mean dimension score per strategy family. Social-normative and covert families concentrate pressure on normative/emotional dimensions; benign dialogues are uniformly low.

TABLE XV
T3 intensity prediction from the dimension-conditioned pipeline on the full release (n=3,432), with the first-release (n=634) values for comparison.

Metric	Full	First release
Spearman ρ	0.651	0.665
Pearson r	0.677	0.691
Quadratic weighted kappa	0.595	0.623
Accuracy within ± 1	0.692	0.722
Exact match	0.216	0.219

first-release and full-scale level. Second, cross-platform variation in norms and politeness may limit transferability; the cross-domain evaluation and a planned cross-lingual extension bound this risk. Third, annotation of covert strategies such as gaslighting is inherently subjective; our two-annotator study (Section IV-E) shows deceptiveness (D_5) is the least reliable dimension, which we report rather than hide. Fourth, the dataset inherits the sampling biases of the source corpus, documented in the data statement. Finally, the benchmark measures the exertion of agentive force, not its effect: a coercive dialogue the listener resists is still annotated as high-pressure, correct for detection

TABLE XVI
Cross-domain zero-shot transfer of the seven-dimension pipeline. "Standard metric" is each corpus's own metric under the source-task decision rule (intensity $\geq$3); "AUROC" is threshold-free and directly comparable. Sampled subsets: MentalManip n=300, MultiManip n=220, TalkDown n=652, ToxicChat n=300.

Corpus	Standard metric	Zero-shot	AUROC
MentalManip	macro-F1	0.615	0.742
MultiManip	macro-F1	0.347	0.706
TalkDown	F1	0.058	0.729
ToxicChat	AUROC	0.587	0.587

but not evidence of harm. This is a resource and analysis benchmark rather than a claim of a complete theory of coercion; the seven dimensions are meant to be a practical, falsifiable abstraction that future datasets can refine.

J. Discussion

Three outcomes matter beyond the numbers. First, reliable dimension profiles provide a continuous, interpretable measurement of how language exerts pressure—from transparent requests to covert manipulation under one rubric—and locate each strategy family precisely in this space. Second, distinct per-family profiles let system designers select parsers and aggregation modes from a small set of validated configurations rather than re-tuning per task. Third, the profile offers an interpretable audit trail: a detection system can justify a "coercive" verdict with concrete dimension evidence (e.g., high D_3, low D_7), the accountability that binary toxicity scores cannot provide. These implications motivate releasing the gold dimension profiles and parser outputs with the benchmark. The central payoff is a shared intermediate representation that makes comparison, auditing, and error analysis possible across different kinds of pressure.

VI. Conclusion

We presented LINGUAFORCE, a new dataset and unified benchmark for the agentive force of discourse in real-world multi-turn dialogues. The full release contains 3,432 dialogues annotated with dialogue-level intensity and seven universal psychological dimensions, and its reliability is verified through consistency with the source pressure labels. We further specify three benchmark tasks, a two-stage dimension-aware pipeline, and research questions on dimension structure and zero-shot generalization. A zero-shot dimension-conditioned readout reaches binary AUC 0.852 and best F1 0.817 on the full release, exceeding a direct zero-shot judge, and transfers moderately to psychological-manipulation and patronizing-language corpora while remaining distinct from generic toxicity. Future work includes stronger multi-parser evaluation, cross-lingual extension, and studying how agentive force interacts with listener vulnerability and cultural norms.

Appendix A
Ethics and Privacy

The dialogues are de-identified and this work performs re-annotation rather than new collection. The release adds no personal information. The study does not collect behavioral or demographic data; the manual checks use already-anonymized text. Annotators may skip distressing content and take breaks. Because the taxonomy could be misused to craft manipulative language, the release is intended for research on detection and analysis, not automated moderation, surveillance, or consequential decisions without human review. The current data are English and drawn from everyday scenarios, so cultural and cross-lingual validity remains an open question. A datasheet documents composition, preprocessing, intended uses, and limitations.

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